Lab experiment

Map Colouring (Constraint Satisfaction Problem)

Coding:

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# Map Colouring Problem using CSP

# Map represented as a graph
graph = {
    'A': ['B', 'C'],
    'B': ['A', 'C', 'D'],
    'C': ['A', 'B', 'D'],
    'D': ['B', 'C']
}

# Available colours
colors = ['Red', 'Green', 'Blue']

# Store colour assignments
assignment = {}

# Check whether colour is valid
def is_safe(region, color):
    for neighbor in graph[region]:
        if neighbor in assignment:
            if assignment[neighbor] == color:
                return False

    return True

# Backtracking function
def map_coloring(regions):
    if not regions:
        return True

    region = regions[0]

    for color in colors:
        if is_safe(region, color):
            assignment[region] = color

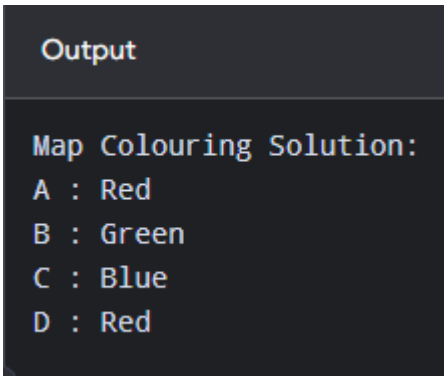
            if map_coloring(regions[1:]):
```

```
                return True

            del assignment[region]

    return False

# Main program
if map_coloring(list(graph.keys())):
    print("Map Colouring Solution:")

    for region, color in assignment.items():
        print(region, ":", color)
else:
    print("No solution exists")
```

Output: